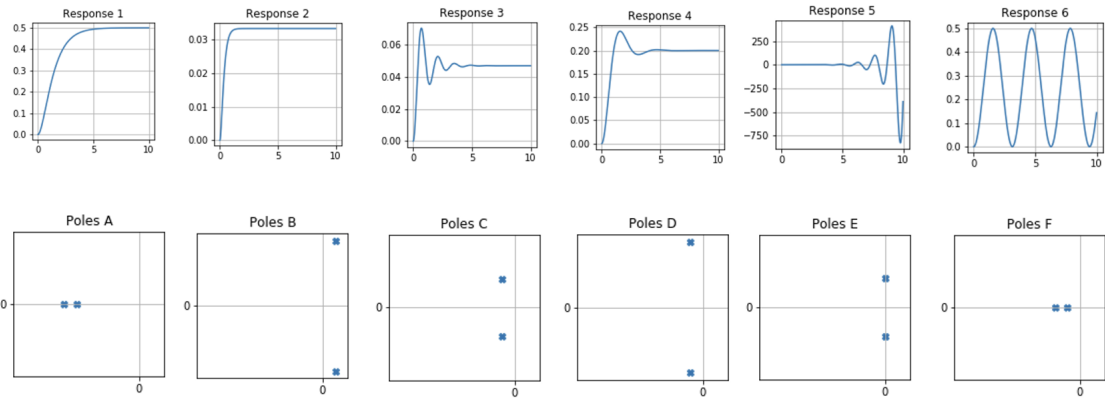


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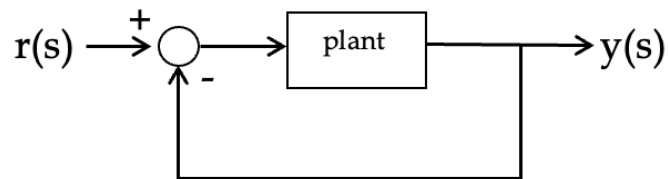
Problem Set 2.1 – Transfer Functions and System I.D.

Question 1

Each of the following plots 1–6 represents the **impulse response** for a second-order system, while the plots A–F represent their **pole locations**. Match each response to the corresponding pole plot.



Question 2



Each of the following

1. $G_1(s) = \frac{1}{s+2}$
2. $G_2(s) = \frac{s+1}{(s+5)(s+3)}$
3. $G_3(s) = \frac{1}{s(s+4)}$

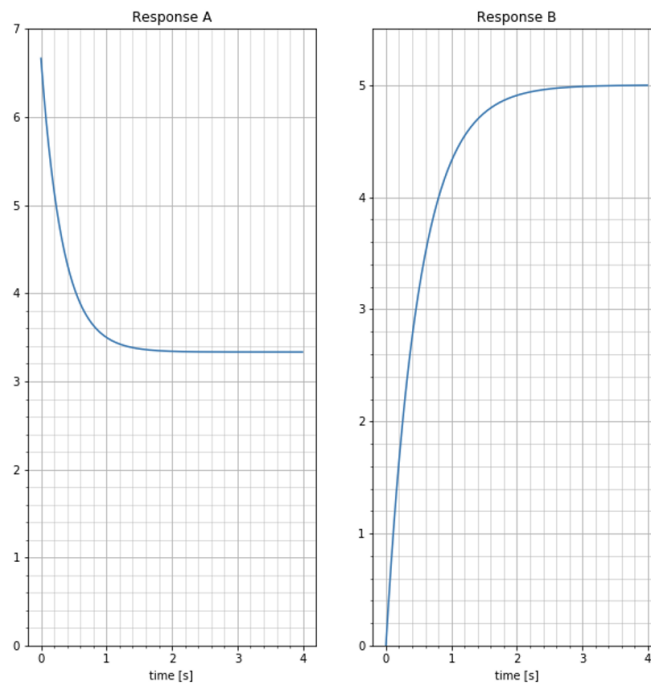
is placed in closed loop, according to the diagram above. For each:

- (a) Calculate the **closed-loop transfer function**.

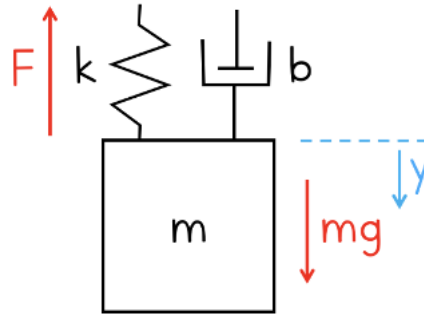
- (b) What is the **type number** of the system? Based on this, state whether you expect the system to track a unit step without error, and calculate the error constant for this input. If the error is non-zero, suggest a **modification** to the system that would lead to perfect tracking.
- (c) For the **open-loop and closed-loop** transfer functions, evaluate the **gain** of the system, and find the **initial** and **final** values of the unit-step response (use the initial- and final-value theorems where applicable).

Question 3

The plots below show the **step responses** obtained from two unknown **first-order** systems. Suggest transfer functions for the systems based on these plots.



Question 4



A dumbwaiter is a small lift for carrying goods (usually food) between floors in a building. A force diagram is shown to the right. The input to the system is the motor force F , and the output is the vertical position y . The mass of the empty lift is $m = 5$ kg, and it can carry a maximum of 45 kg. The cables have spring constant $k = 100$ N/m. A damper with coefficient b moderates the speed. *Hint: pay attention to directions. You only need to consider downward motion.*

- Derive a transfer function relating **position** to **force**, $G(s) = \frac{Y(s)}{F(s)}$. Draw a block diagram depicting the force, position, gravity, and plant.
- It is vital that the system does **not oscillate**, so food does not get spilled. What is the **smallest damping coefficient** that would prevent oscillation, regardless of how full the lift is?
- Using this damping value, **plot** the response of the system to a **unit-step** input in force.
- How would you **modify the system** to track position setpoints accurately?
- How would you modify the system to **counteract gravity**?
- How would you modify the system so the user could specify a constant **velocity** for it to travel at?

Question 5

For each of the following third-order systems,

1. $G_a(s) = \frac{1}{(s+1)(s+10)(s+12)}$

2. $G_b(s) = \frac{s+2.1}{(s+2)(s+5)(s+10)}$

3. $G_c(s) = \frac{1}{(s+5)(s+5.5)(s+7)}$

- (a) Decompose the system into its **modes** and calculate the **residuals** (partial-fraction expansion) for each mode.
- (b) Suggest **second-order and first-order** models to approximate the behaviour of the system (justify dominant-pole/grouping choices).
- (c) **Plot** the step responses for the original model and your first- and second-order approximations, and decide which is the **lowest-order** version that is usable in practice.